Predicting port congestion and vessel waiting delays

Technical report | 21 October 2026 | Synthetic-data prototype

Prediction: at a vessel's arrival, estimate the probability that it waits **more than six hours** before getting a berth. The output includes a selected-outcome confidence score and an explanation. The six-hour cutoff is a prototype assumption.

1. Scope and operational meaning

A record represents one visit to a fictional port. Arrival is entry into the designated waiting area; berth start is the start of berth occupancy. The binary label is 1 when berth start minus arrival exceeds six hours, and 0 otherwise, including equality. This predicts port waiting rather than total shipment transit time. Planners could use alerts to review berth assignments and communicate downstream disruption.

2. Data and simulated ground truth

The exported data contains **500 visits**, of which **98 (19.6%)** are delayed. The assignment permits synthetic data. Seed 42 and 150 warmup visits produce reproducible queue states and historical statistics before the 500 exported arrivals.

Fictional port	Operational berths	Base mean service
PORT_A	3	7.0 hours
PORT_B	5	8.0 hours
PORT_C	7	8.5 hours

Interarrival times are exponential, with an approximate sampling rate of 1.35/hour scaled by 1.45 in daytime and 0.72 otherwise. Port probabilities follow berth counts. Weather persists for 12-hour blocks (severity 0/1/2 with probabilities 0.70/0.23/0.07). Service is lognormal (log-SD 0.45), multiplied by 1 +0.25 times arrival weather severity; 4% of visits incur an additional hidden 4-16-hour disruption.

A first-come-first-served berth heap sets start =max(arrival, earliest berth availability). Sampled service durations and future events are hidden from model inputs. Historical averages use only completed earlier calls, with the configured port mean as a startup fallback. After labelling, each weather/history observation is independently masked with probability 3% (14 weather and 15 history values in this realization).

3. Features known at arrival

Queue per berth; fraction of berths occupied; arrivals in the previous six hours per berth; current weather severity; and average completed service time. Normalization expresses pressure relative to capacity. Visit/port identifiers, final wait, berth-start time, final service duration, and future weather are excluded. Fixed capacities and interchangeable berths simplify real operations.

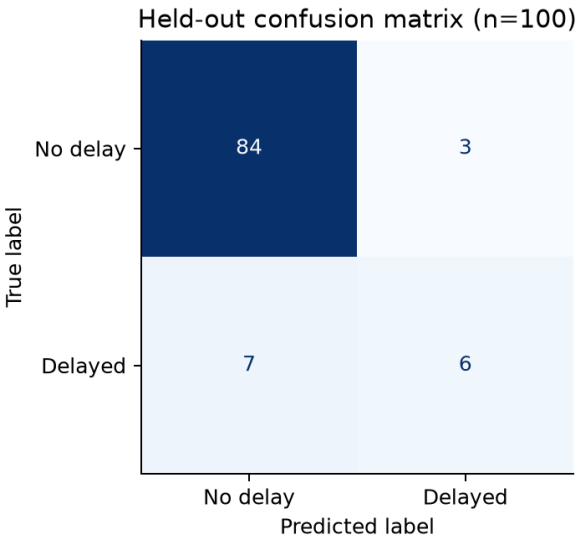
Evaluation and probability quality

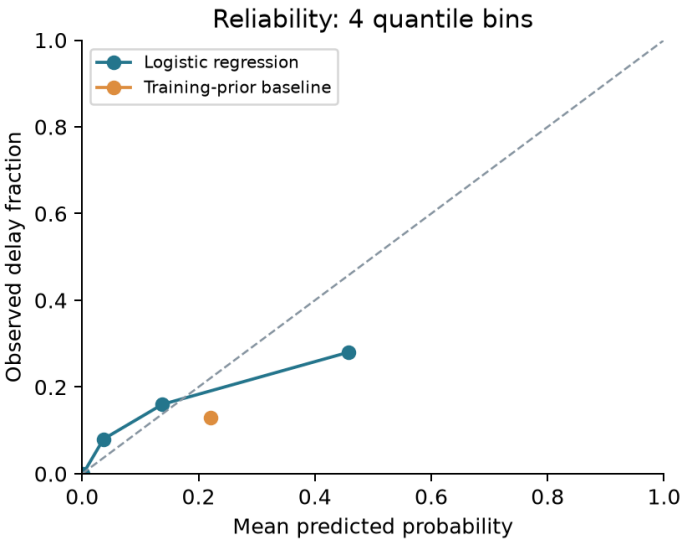
4. Chronological design and model choice

Nominal training/validation/test sizes are 350/50/100 arrivals in time order. A short-wait label becomes known at berth start; a positive label becomes known just after six hours of observed waiting. Removing labels unresolved at the next partition's start leaves **348/47/100** usable visits (2 training and 3 validation exclusions). Only training records fit median imputation and standardization.

Compare a training-prior DummyClassifier with logistic regression. Choose C from 0.1, 1, and 10 using the lowest validation Brier score; selected C is **10**. The classification threshold remains 0.5. No test-based tuning, oversampling, or final refit on test observations is performed.

Held-out metric	Prior baseline	Logistic regression
Accuracy	87.0%	90.0%
Precision / recall	0.0% / 0.0%	66.7% / 46.2%
F1 / ROC-AUC	0.000 / 0.500	0.545 / 0.770
Brier score (lower better)	0.1214	0.1006





Error interpretation: among 100 test visits, the model detects 6 of 13 delays, misses 7, and raises 3 false alarms; 84 non-delays are correctly classified. Accuracy rises only three percentage points over the no-delay baseline. The 46.2% delay recall is a material limitation for disruption prevention. The small validation set has only seven positives; its stronger ROC-AUC (0.993) and F1 (0.833) do not carry through to this later test period.

5. What confidence means

The model estimates p =P(wait >6h). The confidence score equals p for a delayed prediction and 1-p for a non-delayed prediction. It is the selected outcome's model probability, not an independent reliability score or measured accuracy. Four quantile reliability bins compare estimated probabilities and observed fractions. Brier score combines several properties and does not establish calibration alone. No additional calibrator is fitted to the small validation sample. Extreme probabilities and real-port reliability remain unverified.

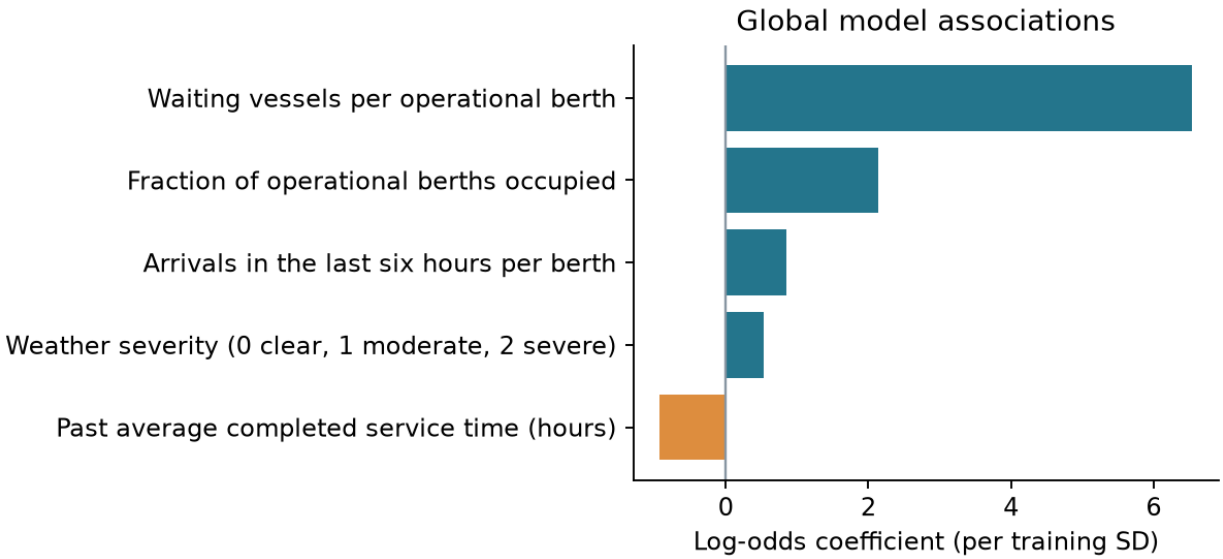
Explanations and operational use

6. Exact model explanations

For each standardized input z, contribution =coefficient times z. Contributions plus the intercept reconstruct the model's log-odds; applying the logistic function returns the prediction. A positive term raises delay risk relative to the training-mean reference, and a negative term lowers it. These are associations within this model, not causal effects or percentage-point changes.

Held-out illustration	Low predicted risk	High predicted risk
Visit / outcome	VISIT_0432 / not delayed	VISIT_0469 / delayed
Estimated delay probability	0.00067%	74.06%
Queue / occupied / capacity	0 / 0 / 5	4 / 5 / 5

These illustrations were selected as the lowest and highest test probabilities after evaluation; they did not select model settings and are not evidence of typical prediction quality. Queue pressure and berth occupancy dominate these particular explanations. Missing features are explicitly flagged when filled with training medians.



7. Actions, limitations, and next steps

Use: a high-risk alert prompts review of berth scheduling and notifications to downstream planners. A low-risk score supports continued monitoring. Outputs are advisory; the model has not demonstrated that any intervention will reduce delay.

Limits: synthetic labels validate the simulator-to-model workflow. The short date range, three seen ports, fixed capacities, simplified weather, and small number of positive outcomes do not establish real-port accuracy, seasonal effects, generalization to new ports, or dependable probabilities. Hidden remaining service times and disruptions create irreducible uncertainty for the chosen inputs.

Next: collect real arrival/berth events and live queue snapshots. Validate at later time periods and unseen ports. Choose alert thresholds against explicit missed-delay and false-alarm costs using new validation periods, then evaluate on a fresh forward holdout.

Reproduction and references

Run python port_delay.py train; python -m unittest discover -s tests -v. The executed notebook, generated CSV, fitted pipeline, metrics, charts, example JSON inputs, and technical report are included locally. Requirements pin the direct packages.

Primary documentation: scikit-learn logistic regression; probability calibration; temporal evaluation. Assignment brief: Aiqon AI Internship Take-Home Assignment (3 Days).